CONTROL ENGINEERING

1. a) To begin with note that since T_0 is constant then $\dot{x}_i = \dot{T}_i - \dot{T}_0 = \dot{T}_i$. In addition, $T_i - T_j$, with $i \neq j$ and $i \in \{1,2,3\}$ and $j \in \{1,2,3\}$, is such that $T_i - T_j = x_i - x_j$. As a result, the state space description of the system, with the indicated state variables, is given by

$$\dot{x} = \begin{bmatrix} -\frac{k_{01}}{\kappa_1} - \frac{k_{12}}{\kappa_1} & \frac{k_{12}}{\kappa_1} & 0 \\ \frac{k_{12}}{\kappa_2} & -\frac{k_{12}}{\kappa_2} - \frac{k_{23}}{\kappa_2} & \frac{k_{23}}{\kappa_2} \\ 0 & \frac{k_{23}}{\kappa_3} & -\frac{k_{23}}{\kappa_3} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\kappa_3} \end{bmatrix} u,$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x.$$

[6 marks]

b) With the indicated values one has

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -1-k_{23} & k_{23} \\ 0 & k_{23} & -k_{23} \end{bmatrix}.$$

i) The characteristic polynomial of the matrix A is

$$p(\lambda) = \lambda^3 + (2k_{23}+3)\lambda^2 + (5k_{23}+1)\lambda + k_{23}.$$

By Routh test one has that the system is asymptotically stable for all $k_{23} > 0$. For $k_{23} = 0$ the characteristic polynomial has one root at $\lambda = 0$ and two roots with negative real part, hence the system is stable.

[4 marks]

ii) The observability matrix is

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -3-k_{23} & k_{23} \end{bmatrix}.$$

This has rank three for all $k_{23} > 0$, hence the system is observable for all $k_{23} > 0$.

[4 marks]

c) For $k_{23} = 0$ one has

$$\begin{aligned} \kappa_1 \dot{T}_1 &= -k_{01}(T_1 - T_0) + k_{12}(T_2 - T_1), \\ \kappa_2 \dot{T}_2 &= -k_{12}(T_2 - T_1), \\ \kappa_3 \dot{T}_3 &= u, \\ y &= T_1 - T_0. \end{aligned}$$

From the equations one observes that the control acts only the the $\dot{T}_3$ equations and the state T_3 does not contribute to the first two equations, while the output

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y can *extract* information only on the states T_1 and T_2. The system is therefore decomposed into a reachable and unobservable subsystem, with state T_3, and an observable and unreachable subsystem, with state (T_1, T_2). The unreachable modes are the roots of the polynomial

$$\lambda^2 + \frac{k_{01}\kappa_2 + k_{12}\kappa_1 + k_{12}\kappa_2}{\kappa_1\kappa_2}\lambda + \frac{k_{01}k_{12}}{\kappa_1\kappa_2},$$

which have negative real part, whereas the unobservable mode is $\lambda = 0$. The system is therefore stabilizable, but not detectable. [6 marks]

Answers

2. a) The resulting closed-loop system is described by the equations

$$\sigma x = (A+BK)x + Bv, \qquad y = Cx.$$

The matrix Z_{SF} which characterizes the zeros of this closed-loop system is

$$Z_{SF}(\zeta) = \begin{bmatrix} \zeta I - (A+BK) & B \\ C & 0 \end{bmatrix}$$

and this can be rewritten as

$$Z_{SF}(\zeta) = \begin{bmatrix} \zeta I - A & B \\ C & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ -K & 1 \end{bmatrix},$$

that is as the product of the matrix characterizing the zeros of the open-loop system and a constant nonsingular matrix. As a result $\det Z_{SF}(\zeta) = \det Z(\zeta)$, which proves that state feedback does not alter the zeros of the system, that is zeros are invariant with respect to feedback action. [8 marks]

b) The equations of the system with output injection are

$$\sigma x = (A+LC)x + Bu, \qquad y = Cx.$$

The matrix Z_{OI} which characterizes the zeros of the system with output injection is

$$Z_{OI}(\zeta) = \begin{bmatrix} \zeta I - (A+LC) & B \\ C & 0 \end{bmatrix}$$

and this can be rewritten as

$$Z_{OI}(\zeta) = \begin{bmatrix} I & -L \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \zeta I - A & B \\ C & 0 \end{bmatrix},$$

that is as the product of the matrix characterizing the zeros of the open-loop system and a constant nonsingular matrix. As a result $\det Z_{OI}(\zeta) = \det Z(\zeta)$, which proves that output injection does not alter the zeros of the system, that is zeros are invariant with respect to output injection. [6 marks]

c) The zeros of the open-loop system are the complex numbers such that

$$\det \begin{bmatrix} \zeta - 1 & 1 \\ 1 & 0 \end{bmatrix} = 0,$$

which shows that the open-loop system does not have any zero. For the system with feedforward one has

$$\det \begin{bmatrix} \zeta - 1 & 1 \\ 1 & -\delta \end{bmatrix} = -\delta(\zeta - 1) - 1,$$

which shows that the system with feedforward has a zero at $\zeta = -\dfrac{1}{\delta} + 1$. When $\delta \to 0$, the zero tends to infinity, and this *recovers* the fact that for $\delta = 0$ the system does not have any zero. In conclusions, zeros are not invariant with respect to feedforward action. [6 marks]

Answers

3. a) The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} 1 & 1 \\ 1 & 2+a \end{bmatrix}$$

and its determinant is $\det\mathscr{R} = 1 + a$. The system is therefore reachable for all $a \neq -1$. For $a = -1$ the matrix A has eigenvalues 1 and -1, both on the boundary of the unity disk. As a result, the system is not controllable nor stabilizable. [4 marks]

b) For $a = 0$ the system is reachable, hence it is possible to reach any final state from any initial state in at most $n = 2$ steps. Considering the given initial condition one has

$$x_1(1) = 1 + u(0), \qquad x_2(1) = 2 + u(0),$$

and

$$x_1(2) = 1 + u(0) + u(1), \qquad x_2(2) = 2 + 2u(0) + u(1).$$

As a result, the final state $x_1(2) = x_2(2) = 2$ cannot be reached in one step, but it can be reached in two steps selecting $u(0) = -1$ and $u(1) = 2$. [6 marks]

c) Let $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$ and note that

$$A + BK = \begin{bmatrix} k_1 + 1 & k_2 \\ k_1 + 2 & k_2 + a \end{bmatrix}.$$

The characteristic polynomial of the matrix $A + BK$ is

$$p(\lambda) = \lambda^2 - (1 + a + k_1 + k_2)\lambda + (a - k_2 + ak_1).$$

To have both closed-loop eigenvalues at zero one has to solve the equations

$$1 + a + k_1 + k_2 = 0, \qquad a - k_2 + ak_1 = 0,$$

for k_1 and k_2, yielding

$$k_1 = -\frac{2a+1}{a+1} \qquad k_2 = -\frac{a^2}{a+1}.$$

The state feedback gain is well-defined for all $a \neq -1$, which is consistent with the fact that for $a = -1$ the system is not reachable and the unreachable mode is not at 0, that is the system is not controllable. [6 marks]

d) For $a = -1$ the reachability matrix is

$$\mathscr{R} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

which has rank one. The set of reachable states is

$$\text{Im}\,\mathscr{R} = \text{span}\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

that is all states on the line $x_1 = x_2$. [4 marks]

Answers

4. a) The observability matrix of the system is

$$\mathcal{O} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

which has rank three, hence the system is observable. [2 marks]

b) Since d is measurable one could use it in the observer equations. These can be defined as

$$\begin{aligned} \dot{\xi}_1 &= \xi_2 + l_1(\hat{y} - y), \\ \dot{\xi}_2 &= \xi_3 + l_2(\hat{y} - y), \\ \dot{\xi}_3 &= d + l_3(\hat{y} - y), \end{aligned}$$

with $\hat{y} = \xi_1 + \xi_2 + \xi_3$, resulting in the state estimation error equations (in which $e_i = x_i - \xi_i$)

$$\dot{e}_1 = e_2 + l_1(e_1 + e_2 + e_3), \qquad \dot{e}_2 = e_3 + l_2(e_1 + e_2 + e_3), \qquad \dot{e}_3 = l_3(e_1 + e_2 + e_3).$$

The "A" matrix of the estimation error system is

$$A_e = \begin{bmatrix} l_1 & l_1 + 1 & l_1 \\ l_2 & l_2 & l_2 + 1 \\ l_3 & l_3 & l_3 \end{bmatrix}$$

and its characteristic polynomial is

$$p(\lambda) = \lambda^3 - (l_1 + l_2 + l_3)\lambda^2 - (l_2 + l_3)\lambda - l_3.$$

This has roots -1, -2 and -3 provided $l_1 = 5$, $l_2 = -5$ and $l_3 = -6$. [6 marks]

c) i) The matrices describing the problem are

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \qquad H = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

[4 marks]

ii) Let

$$N = \begin{bmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{bmatrix}, \qquad F = \begin{bmatrix} f_1 & f_2 \end{bmatrix}.$$

A direct computation shows that $VD = 0$ and that

$$NV - VA + LC = \begin{bmatrix} n_{11} & n_{12} - 1 & 0 \\ n_{21} + 1 & n_{22} + 1 & 0 \end{bmatrix}.$$

Hence selecting

$$N = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

yields $NV - VA + LC = 0$. [4 marks]

iii) The variable e is defined as

$$e = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

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with $e_1 = z_1 - x_1$ and $e_2 = z_2 - x_2$. A direct calculation yields

$$\dot{e}_1 = e_2 \qquad \dot{e}_2 = -e_1 - e_2,$$

that is $\dot{e} = Ne$. Note that the matrix N has both eigenvalues with negative real parts, hence e converges exponentially to zero (the error system is exponentially stable). [2 marks]

iv) Since e_1 and e_2 converge exponentially to zero, z_1 and z_2 are exponentially converging estimates of x_1 and x_2, respectively. The variable $\hat{w}$ provides, therefore, an estimate of x_1 if $f_1 = 1$ and $f_2 = 0$. [2 marks]

Answers